

Identification of most predictive array CGH features of breast cancer subtype.

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Abstract

Breast cancer is the most common cancer diagnosis for women worldwide. The disease is highly heterogeneous, with different subtypes having different prognoses and therefore requiring tailored treatment plans. Historically, the inability to accurately identify subtypes early after diagnosis has led to the overtreatment of many breast cancer patients, with many of them undergoing unnecessary chemotherapy and suffering the highly damaging side effects of that treatment. Recent research efforts have focused on developing methods for the early identification of breast cancer subtypes so that patients can receive tailored, precise treatments. In this study, we develop and test multiple machine learning classifiers for the task of breast cancer subtype prediction based on array CGH data, we select the 3 best performing classifiers and combine them into a single soft voting ensemble classifier for optimal results. Our aim is to identify the array CGH features that are most predictive of breast cancer subtype and to assess which machine learning approaches are most suitable for this task. Our results show that logistic regression with elastic nets regularization had the best performance, followed by the support vector machines and random forest classifiers. Furthermore, we identify several key chromosomal regions that are highly predictive of breast cancer subtype across the different classifiers.

Introduction

Breast cancer is the most commonly diagnosed cancer across women worldwide, with an estimated 2.3 million new cases and 670.000 deaths each year (World Health Organization [WHO], 2024). Because the disease is highly heterogeneous, being able to classify tumors in their early stages is essential for choosing the right treatment for the patient (Cardoso et al, 2016). Currently, tumors are classified in the clinic based on the presence or absence of three different receptors: Estrogen Receptor (ER), Progesterone Receptor (PR), and Human Epidermal growth factor Receptor 2 (HER2). This results in the categorizing of breast cancer in three main subtypes, which are HR+, HER2+ and Triple Negative. In each of these subtypes the expression or amplification of these receptors differ, and thus each subtype requires a different treatment plan.

It has been shown that relying only on standard clinical and histological parameters is often not enough to capture the biological diversity of the tumors and to accurately predict the tumors in early stages. The lack of accurate classification has led to significant overtreatment in the past, leading to a lot of patients receiving chemotherapy which was not actually needed for treatment ('t Veer et al, 2002). The study by 't Veer et al. (2002) proved that a comprehensive genomic signature of the patient could predict a patient's risk of distant metastasis far better than previous clinical options. In this study we want to further delve into the problem of tumor classification and expand on these previous findings.

Studies like 't Veer et al. (2002) successfully used high-dimensional gene expression (mRNA) arrays for the breast cancer subtype classification. But in this study we used array CGH data. aCGH data measures genome-wide DNA copy number variations across the entire genome per position by comparing a sample against a reference (Theisen, 2008). It shows for each position whether there is a loss, no change (normal), gain, or amplification for that variant. This results in high resolution genome wide screening of genomic copy number variations (CNVs), and is a good method to look for differences in the copies of the DNA across the genome (Shinawi & Cheung, 2008). The chromosomal aberrations which are shown by breast cancer are highly informative to distinguish the different subtypes (Chin et al., 2006). In this study we used a dataset consisting of 100 breast cancer samples distributed from the three main subtypes. For each sample, chromosomal DNA quantities were measured using a high-resolution array CGH platform with 244,000 distinct genomic probes. Before our study, the raw array CGH data was preprocessed to convert the continuous chromosomal DNA quantity measurements into discrete sections for each of the 244,000 regions. The data was categorized into four states: loss (-1), normal (0), gain (1), and amplification (2). Furthermore dimension reduction has been done, by combining stretches with identical CNV profiles.

Using this data can lead to several problems. Since our data consists of 244,000 features, but only 100 patient samples, this is really high dimensional data. Feeding this data into a classifier will cause overfitting and other problems. To prevent these problems rigid feature selection is necessary to find the most predictive chromosomal regions (Haury et al, 2011). Furthermore in such high dimensional data a strict double-loop cross-validation scheme is necessary to prevent data leakage and selection bias (Wessels et al. 2005). This pre-precossing and preparations before the use of actual classifiers is essential in preventing

bias in high dimensional data (Greener et al 2022). Previous bioinformatics approaches have shown that feature selection combined with classifiers, such as Random Forests, are effective at classifying high-dimensional cancer genomic data (Guyon et al., 2002), and our goal is to find and build the best classifier for our data.

This leads to our first research question; Which array CGH features are most predictive of breast cancer subtype? Based on the predicted cancer subtype, a prognosis could be made and a treatment plan could be formulated. To optimize prediction, we will select features and then evaluate the performance of several classifiers. This leads us to our second research question: which classifier method predicts breast cancer subtype best?

To answer these questions we developed and optimized a machine learning model. This is done by firstly performing data exploration on our data set, then using logistic regression (elastic net) for feature selection to reduce the amount of features and find the relevant ones. Then we evaluated the performance of five different classifiers (Logistic Regression, Random Forest, SVM, KNN, and Neural Networks) within a nested cross-validation scheme to find the best classifier, and then finally optimize that classifier to get the final model.

Methods

The data consists of 2834 features across 100 samples. Each sample is labeled as one of three breastcancer subtypes: HER2+, HR+ or Triple Negative. In the dataset there were 32 samples with subtype HER2+, 36 samples with subtype HR+ and 32 samples with subtype Triple Negative. The features are expression of chromosomal regions, values indicating below or above average expression [-1, 0, 1, 2]. Need to add to this, but need access to canvas for that.

The features were first filtered based on correlation. Features with >95% correlation with another feature were removed from the dataset. To select features to train the model on, the top 30 overall most predictive features were included. These were determined using a multi-class ANOVA F-test. In order to assure the most distinguishing features were included as well, the top 5 most predictive features unique to each subtype (not already included in the overall top 30) were also included in the dataset. These were determined using Binary One-vs-Rest ANOVA profiles. This resulted in a dataset of 45 features.

In order to determine the most accurate model, 5 different types of classifiers were trained on this dataset. These classifiers were Logistic Regression with elastic net penalisation, Random Forest, Neural Network with two hidden layers, Support Vector Machine (SVM) and k-nearest neighbour. The top three best performing models were then combined into a single soft voting ensemble classifier.

For purposes of preventing overfitting, a repeated stratified nested cross-validation scheme was used for all classifiers. This was done for the initial training of the models for comparison, as well as in the final ensemble classifier. A 5 fold CV was done across 5 independent random shuffles of the data. For each split, an inner 3-fold CV was done to optimize model hyperparameters. The 5 fold outer CV and 3 fold inner CV were chosen based on the available number of samples, and ensured that enough data was available for

training and testing. A graphical representation of the CV can be found in figure 1.

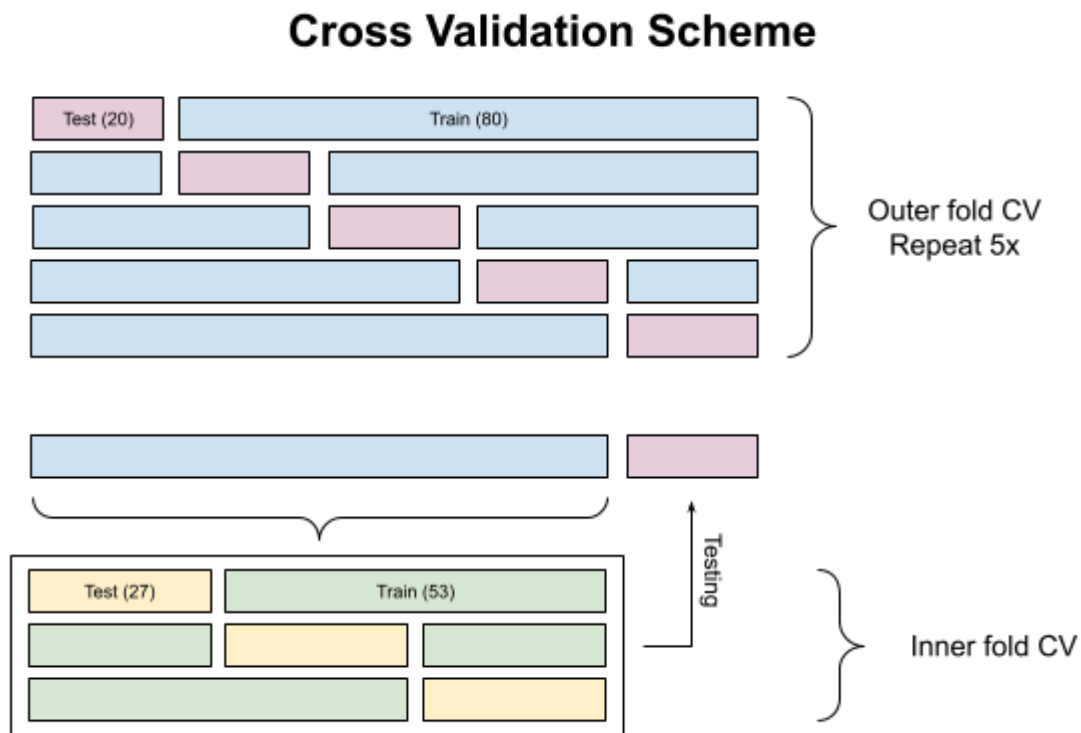


Figure 1: Graphical representation of the nested cross validation scheme.

Tuning the hyperparameters was done in the inner loop of the nested cross validation to prevent data leakage. Finally, to ensure the model did not overfit to the majority class, `'class_weight=balanced'` was set in the classifier, penalising more heavily if a rarer subtype is misclassified.

The final **soft voting** ensemble classifier consisted of three classifiers. The elastic net penalized logistic regression, the random forest classifier and the support vector machine. The three classifiers provided a prediction with a confidence score for each new sample. This was then averaged, resulting in a final prediction.

This model was used to make predictions for a validation dataset. The correct labels of this dataset are currently unknown, but will help determine the final accuracy of this classifier.

Results

Five machine learning classifiers were evaluated using repeated nested cross-validation. Elastic Net Logistic Regression, Random Forest, Support Vector Machine (SVM), Neural Network, and K-Nearest Neighbour (KNN). Model performance was assessed using mean classification accuracy, precision, recall, and F1-score.

Elastic Net Logistic Regression achieved the highest overall performance, with a mean accuracy of $0.860 (\pm 0.020)$. The model demonstrated excellent classification of HER2+ tumours, achieving a precision and recall of 0.97 for this subtype. Performance was lower for HR+ and Triple Negative tumours, with F1-scores of 0.82 and 0.80, respectively.

The SVM classifier achieved an accuracy of $0.830 (\pm 0.060)$, while Random Forest achieved $0.820 (\pm 0.024)$. Both models also showed near-perfect classification of HER2+ samples, but reduced performance in distinguishing HR+ from Triple Negative tumours. Neural Networks achieved a lower average accuracy of $0.750 (\pm 0.089)$, showing substantial variability between folds. KNN performed worst overall, achieving an accuracy of 0.660.

Model	Accuracy
Elastic Net Logistic Regression	0.860 ± 0.020
Support Vector Machine	0.830 ± 0.060
Random Forest	0.820 ± 0.024
Neural Network	0.750 ± 0.089
K-Nearest Neighbour	0.660

Table 1: Used Models and corresponding accuracy

Feature selection was performed using multi-class ANOVA F-test to identify the 30 most predictive genomic regions. To better find subtype-specific genomic alterations, the top 5 unique features from One-vs-Rest ANOVA analyses for each subtype were added. This resulted in a set of 45 features.

To further improve generalizability and prevent data leakage, all models were evaluated using repeated nested cross-validation consisting of 5 outer folds across 5 independent random shuffles, with an inner 3-fold cross-validation loop for hyperparameter optimization. Class balancing was also applied during training to reduce bias toward more frequent classes. These modifications substantially improved model stability and reduced overfitting compared to initial exploratory models.

HER2+ tumours were classified with very high accuracy across nearly all models. Precision and recall values approaching 1.00 for the best performing classifiers. Most misclassifications occurred between HR+ and Triple Negative tumours, indicating greater overlap in their genomic copy number variation profiles.

The three best performing classifiers were combined into a soft voting ensemble classifier. These classifiers were Elastic Net Logistic Regression, Random Forest, and SVM. This ensemble averaged prediction probabilities from the individual models to generate a final consensus prediction and confidence score. The ensemble achieved a consensus accuracy of 0.830 across 25 repeated nested cross-validation runs. Although the ensemble did not

outperform logistic regression alone, it produced more stable predictions and provided confidence estimates for each sample classification.

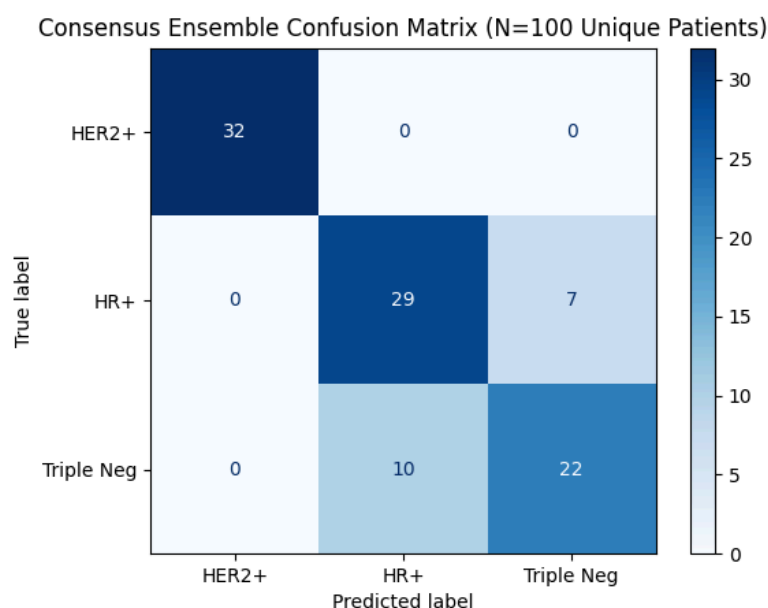


Figure 2: Confusion matrix of the final model

The selected predictive genomic regions were biologically relevant and strongly associated with known breast cancer mechanisms. The most predictive feature corresponded to the chr17:35.07-35.28 Mb region containing the ERBB2 (HER2) locus, a defining characteristic of HER2-positive breast cancer. Additional highly ranked regions included loci containing GRB7 and TOP2A, both frequently co-amplified with ERBB2. These findings indicate that the feature selection process successfully identified relevant genomic alterations associated with breast cancer subtype biology.

Finally, Elastic Net Regression was identified as the best performing classifier for this task. Therefore, it represents the most suitable biomarker based classification approach.

Interpretation of Results

The results of our study show that the elastic nets linear regression classifier performs best on the task of breast cancer subtype prediction, followed closely by the support vector machines and random forest classifiers. Neural networks perform worse but still have a better than random performance, and k-nearest neighbors have the worst performance overall.

The poor performance of the k-nearest neighbors classifier is not surprising, given that the k-nearest neighbor algorithm is rule-based and involves no actual learning of the relationship between the features and target variable, making it unlikely to be suitable for tasks involving complex biological data such as cancer subtype classification.

The fact that the neural networks model performed worse on this task likely has to do with the chosen network architecture. Neural networks can be very powerful models when the right architecture is used, and deep learning approaches are currently showing a lot of promise in their application to biological data, a notable example is protein structure prediction with alphafold (Jumper et al., 2021). However, the network we used for this task had only two hidden layers, limiting its ability to learn a meaningful relationship between the features and target variable. Building the right neural network architecture requires domain expertise with the area of application, and interpreting these models is challenging, making their usefulness limited for our research question which requires good interpretability, as we want to not only predict breast cancer subtypes but also to identify the important features that drive this prediction.

The good performance of the linear regression classifier on the task suggests that the 3 breast cancer subtypes are linearly separable in the feature space of array CGH data. This suggests that the relationship between DNA copy numbers and breast cancer subtypes can be linearly approximated and therefore satisfactory prediction performance is achieved without requiring complex non-linear learning algorithms, such as deep learning architectures. This conclusion is further supported by the fact that the nonlinear random forest classifier, although it has comparable performance, does not outperform the regularized linear regression classifier on the data. Another possibility is that the logistic regression model is overfitting.

Include some discussion about the soft voting ensemble classifier.

Furthermore, we identified the array CGH features that are most predictive of breast cancer subtype across the different classifiers. We found that, as would be expected, these features correspond to DNA regions that code for genes associated with the phenotypes of the three different subtypes, for example genes coding for the receptor types that characterize each of subtypes and genes that co-vary with them. A notable example is the chr17:35.07–35.28 Mb region which corresponds to the ERBB2 locus coding for Human Epidermal growth factor Receptor 2 (HER2) and the chr17:34.99–35.07 Mb region corresponding to the GRB7 gene coding for Growth Factor Receptor-Bound Protein 7, which is co-amplified with ERBB2.

Discussion and Conclusion

In conclusion, we built multiple machine learning classifiers for the task of breast cancer subtype prediction based on array CGH data, based on their validation performance we selected the top 3 models and combined them in a single soft voting ensemble predictor for optimal results. Our aim was to identify the array CGH features that are most predictive of cancer subtype across classifiers and to assess which machine learning approaches are most suitable for the task. We show that the most important features correspond to DNA regions that code for the receptors characteristic of specific subtypes and genes that co-vary with them. Furthermore, we show that the best performing models are: logistic regression with elastic nets regularization, random forest, and support vector machines, with neural networks having a limited performance and the k-nearest neighbours model performing poorly on the task.

Author contribution statement

Laurie Haj Islam: Building the neural network model, writing of the “abstract”, “interpretation of the results”, and “discussion and conclusion” sections.

Lora Lagerweij: Feature selection, Random Forest and Logistic regression model, Final Ensemble model, Methods, predictions of validation dataset.

Jasper Simons: SVM model, Cross validation scheme, Introduction

Stephan Chelaru: K-Nearest Neighbour Model, Writing of the “Results” section.

Note to reviewers

Our logistic regression model might be overfitting, we are going to rerun all models after editing our CV approach, thus the conclusions might change a bit in the final version, specifically the points about the 3 subtypes being linearly separable in the feature space.

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